A15536W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics, and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMIC ANALYSIS

OUTLINE ANSWERS

TRINITY TERM 2022

Saturday 18 June

Opening time 09:30 (BST)

You have 3 hours to complete the paper and up to 30 minutes technical time to upload your answer file.

Mode of completion: Mixed Mode

Please start the answer to each question in the relevant response box. Questions with sub-sections (e.g., Q1a & Q1b) should all be answered in the response box for the corresponding overall question number.

There are 6 questions in this paper.

Answer **four** *questions.*

Candidates may use their own calculators.

1. (a) Consider the vectors $(c, 1, 1)$, $(1, c, 1)$, and $(1, 1, c)$. For which real numbers c do these three vectors fail to form a basis of $\mathbb{R}^3$? Hint: look for more than one. For each value of c you find, what is the dimension of the space spanned by the vectors?

Solution: Check if the determinant of the matrix formed by the vectors is zero.

$$\det \begin{pmatrix} c & 1 & 1 \\ 1 & c & 1 \\ 1 & 1 & c \end{pmatrix} = c^3 - 3c + 2,$$

either by hand (recursively) or by WolframAlpha. This is zero at $c = 1$ and $c = -2$ because $c^3 - 3c + 2 = (c - 1)^2(c + 2)$.

For $c = 1$, the three vectors are identical, $(1, 1, 1)$, they span a one-dimensional subspace: a straight line going through the origin and $(1, 1, 1)$.

For $c = -2$ the three vectors add up to $(0, 0, 0)$, they span a 2-dim plane going through the origin.

[20%]

(b) Find a 2-by-2 matrix A of real numbers such that one of its eigenvalues is $\lambda_1 = 1$ with eigenvector $(1, 2)$, the other is $\lambda_2 = -1$ with eigenvector $(2, 1)$. Consider the dynamical system in continuous time with law of motion $\dot{\mathbf{y}} = A\mathbf{y}$ where $\mathbf{y} = (y_1, y_2)$. Discuss its stability properties.

Solution: Denoting the elements of A as usual, the eigenvalue-eigenvector equations are

$$\begin{aligned} a_{11} + 2a_{12} &= 1 \\ a_{21} + 2a_{22} &= 2 \\ 2a_{11} + a_{12} &= -2 \\ 2a_{21} + a_{22} &= -1 \end{aligned}$$

Multiply the first equation by -2 and add it to the third one to eliminate a_{11} and find $a_{12} = 4/3$. Sub back to find $a_{11} = -5/3$. Similarly, from the second and fourth equations $a_{21} = -4/3$ and $a_{22} = 5/3$. Hence

$$A = \begin{pmatrix} -5/3 & 4/3 \\ -4/3 & 5/3 \end{pmatrix}.$$

Double checked with WolframAlpha, correct.

The dynamical system defined by A is saddle-path stable without needing to derive A because we are told one eigenvalue is positive the other is negative. The stationary point is $(0, 0)$ by homogeneity, the saddle path is defined by the eigenvector associated with the negative eigenvalue, i.e., $(2, 1)$. A very nice solution might give closed form solutions: $y_1 = c_1 e^t + 2c_2 e^{-t}$ and $y_2 = 2c_1 e^t + c_2 e^{-t}$ where c_1 and c_2 are constants determined by e.g. initial conditions.

[30%]

(c) Consider the function

$$f(x,y) = \begin{cases} \frac{3xy}{x^2+2y^2} & \text{if} \quad x \neq 0 \\ 0 & \text{if} \quad x = 0 \end{cases}$$

(i) For any vector $\boldsymbol{v} = (v_1, v_2) \in \mathbb{R}^2$, does the directional derivative of f in the direction of $\boldsymbol{v}$ at point $(0,0)$ exist? If so, find it.

Solution:

$$Df(0,0,\boldsymbol{v}) = \lim_{t \to 0} \frac{f(tv_1, tv_2) - f(0,0)}{t}$$

If $v_1 = 0$ then $Df(0,0,\boldsymbol{v}) = 0$. Otherwise:

$$Df(0,0,\boldsymbol{v}) = \lim_{t \to 0} \frac{t^2 v_1 v_2}{t^2 v_1^2 + 2t^2 v_2^2)} = \lim_{t \to 0} \frac{v_1 v_2}{v_1^2 + 2v_2^2}$$

So the directional derivatives always exist at $(0,0)$.

(ii) Is f differentiable at $(0,0)$?

Solution: No, f is not continuous at $(0,0)$ and hence no differentiable. To see that f is not continuous consider a sequence $\{(x_n, y_n)\}_n = \{(\frac{1}{n}, \frac{1}{n})\}_n$ which converges to $(0,0)$. $f(x_n, y_n) = \frac{3/n^2}{3/n^2} = 1$ which does not converge to $f(0,0) = 0$.

[30%]

(d) Consider the function $g(x,y,z) = z(x+y) - y\ln(z)$.

(i) Verify that for all t, $g(t,-t,1) = 0$. For which values of t does $g(x,y,z) = 0$ define z as an implicit function of (x,y) around $(t,-t)$?

Solution:first question trivial. g is a C^1 function for any $z > 0$ and $g_z(x,y,z) = x + y - y/z$, hence $g_z(t,-t,1) = t$ which is different form 0 whenver $t \neq 0$. Hence, when $t \neq 0$ we can apply the implicit function theorem and z is indeed an implicit function of (x,y) around $(t,-t)$.

(ii) For the values of t found in (i) compute the partial derivatives of z at $(t,-t)$.

Solution: Whenever $t \neq 0$, $\frac{\partial z}{\partial x}(t,-t) = \frac{g_x(t,-t,1)}{g_z(t,-t,1)} = 1/t$ and $\frac{\partial z}{\partial y}(t,-t) = \frac{g_y(t,-t,1)}{g_z(t,-t,1)} = 1/t$

[20%]

2. (a) Consider the following constrained problem:

$$\max f(x,y) \quad \text{subject to} \quad x + 2y \leq 9, x \geq 0, y \geq 0$$

Argue whether the following statements are true or false:

(i) If (x^*, y^*) maximises $L(x,y,\lambda^*) = f(x,y) - \lambda^*(x+2y-9)$ with $\lambda^*(x+2y-9) = 0$

and (x^*, y^*) is feasible, then (x^*, y^*) is a solution of the constrained problem (P).

Solution: False: This is an inequality constrained problem, we need to impose $\lambda \geq 0$. For example: consider $f(x, y) = -x^2 - y^2 + 2x + 2y - 2$ and $\lambda = -1$. Then $(\frac{3}{2}, 2)$ maximises $L(x, y, -1)$ but it is not the solution to (P). (The solutions is $(1, 1)$, which is the unconstrained maximum).

(ii) If f is continuous and (x^*, y^*) is the unique solution to the Kuhn-Tucker first order conditions, then (x^*, y^*) is a global maximum of (P).

Solution: True. A correct explanation should mention two things: First, the constraint qualification is always satisfied, and hence the K-T FOC are necessary for a local maximum. Second, the constrained set is compact and f is continuous, hence by Weierstrass Theorem the global maximum is achieved. Since a global maximum is a local maximum it must satisfy the FOC, and since there is only one solution it must be the global maximum.

(iii) If $f(x, y)$ is quasi-concave and increasing in both x and y then the constraints $x \geq 0$ and $y \geq 0$ never bind.

Solution: False: With quasi-concave, quasi-linear utility the solution might be at a boundary. For example $u(x, y) = 4x^{\frac{1}{2}} + y$. The solution is at $x = 9$ and $y = 0$.

(b) A consumer has a utility function $u(x, y) = 3\sqrt{x} + y$ for the two available goods. Her budget constraint is $x + 2y \leq m$, where $m > 0$ is her income.

(i) Explain briefly why the consumer will always consume a strictly positive amount of x.
Solution: Marginal utility of x goes to infinity as x goes to zero, so it is always optimal to consume a positive amount.

(ii) Write down the Kuhn-Tucker first-order conditions for the consumer and find the solution as a function of m. Is the solution you found optimal?
Solution: Lagrangian:

$$L = 3\sqrt{x} + y - \lambda(x + 2y - m) + \gamma y$$

K-T FOC:

$$\frac{3}{2}x^{-\frac{1}{2}} - \lambda = 0 \tag{1}$$

$$1 - 2\lambda + \gamma = 0 \tag{2}$$

$$\lambda(x + 2y - m) = 0, \quad \lambda \geq 0, \quad x + 2y \leq m \tag{3}$$

$$\gamma y = 0, \gamma \geq 0, y \geq 0 \tag{4}$$

Suppose $y > 0$ so $\gamma = 0$, then (2) implies $\lambda = \frac{1}{2} > 0$ so the budget constraint binds. Moreover, (1) implies $x = 9$ and hence $y = \frac{m-9}{2}$. This is a solution whenever $y \geq 0$ so $m \geq 9$.
If $y = 0$, then by (2) $\lambda > 0$ and the budget constraint binds so $x = m$. Hence by (1) $\lambda = \frac{3}{2}m^{-\frac{1}{2}}$ and by 2) $\gamma = 3m^{-\frac{1}{2}} - 1 \geq 0$ if and only if $m \leq 9$. Therefore there is a unique solution to the K-T FOC:

$$(x, y) = \begin{cases} (9, \frac{m-9}{2}) & if \quad m \geq 9 \\ (m, 0) & if \quad m < 9 \end{cases}$$

This is a concave problem so the solutions to the KT FOC are both necessary and sufficient. So this is indeed the solution to the constrained problem.

The consumer also faces a time constraint $3x + 4y \leq 48$.

(iii) For which values of m is this new constraint binding?
Solution: If $m < 9$ then the time constraint those not bind. If $m > 9$ then $3 * 9 + 4 * \frac{m-9}{2} \leq 48$ if and only if $m \leq 19.5$. If $m > 19.5$ the time constraint if binding. Note that for that if $m > 19.5$, $y > 0$ always so we can ignore the positivity constraint.

(iv) Find the new solution of the constrained problem as a function of m.
Solution: Suppose $m > 19.5$ (if $m < 19.5$ the solution is the one above). The new Lagrangian:

$$L = 3\sqrt{x} + y - \lambda(x + 2y - m) - \mu(3x + 4y - 48)$$

KT FOC:

$$\frac{3}{2}x^{-\frac{1}{2}} - \lambda - 3\mu = 0 \tag{5}$$
$$1 - 2\lambda - 4\mu = 0 \tag{6}$$
$$\lambda(x + 2y - m) = 0, \quad \lambda \geq 0, \quad x + 2y \leq m \tag{7}$$
$$\mu(3x + 4y - 48) = 0, \quad \mu \geq 0, \quad 3x + 4y \leq 48 \tag{8}$$

Suppose the two constraints bind: solving $x + 2y = m$ and $3x + 4y = 48$ simultaneously: $x = 48 - 2m$ and $y = \frac{3m-48}{2}$. Solving (5) and (6) simultaneously, $2\mu = 3x^{-\frac{1}{2}} - 1$ and $2\lambda = 3 - 6x^{-\frac{1}{2}}$. Hence, $\mu > 0$ and $\lambda > 0$ iff $4 < x < 9$, which means $19.5 < m < 22$.

Suppose now that the budget constraint is not binding, so $\lambda = 0$, then $\mu = \frac{1}{4} > 0$ by (6) and $x = 4$ by (5). Finally the time constraint implies $y = 9$.

Hence the overall solution is :

$$(x, y) = \begin{cases} (4, 9) & if \quad m \geq 22 \\ (48 - 2m, \frac{3m-48}{2}) & if \quad 19.5 < m < 22 \\ (9, \frac{m-9}{2}) & if \quad 9 \leq m \leq 19.5 \\ (m, 0) & if \quad m < 9 \end{cases}$$

[60%]

3. Consider the space $X = \mathbb{R}^2_+$ of vectors composed by two positive real numbers. Think of it as bundles (x_1, x_2) describing the amount of two goods consumed. We will be describing choices and preferences of a character called Dunno. We will consider menus A defined by income m and prices (p_1, p_2), where menu A consists of the set of bundles such that $p_1x_1 + p_2x_2 \leq m$.

   (a) Menu A_1 is defined by income 20 and prices $(p_1, p_2) = (1, 1)$, that is, the set of bundles such that $x_1 + x_2 \leq 20$. In this menu, Dunno chose bundle $(10, 10)$. Menu A_2 is defined by income 30 and prices $(p_1, p_2) = (2, 1)$, that is, the set of bundles such that $2x_1 + x_2 \leq 30$. In this menu, Dunno chose bundle $(15, 0)$.

      (i) Discuss whether property α can be applied to these two menus in order to understand whether Dunno's choices are rational.

      Solution: Property α is silent about these two menus because none of them is contained in the other. Notice that $(20, 0)$ belongs to A_1 but not to A_2, while $(0, 30)$ belongs to A_2 but not to A_1.

      (ii) Use the revealed preference of Dunno's choices and strict monotonicity to explain why Dunno's choices seem irrational.

      Solution: Bundle $(10, 10)$ was revealed preferred to bundle $(20, 0)$ in the first menu. Bundle $(20, 0)$ dominates bundle $(15, 0)$ (if students want to use strict monotonicity only over the two quantities, they may identify $(19, 1)$ in the first place). Now, bundle $(15, 0)$ was revealed preferred to bundle $(10, 10)$ in the second menu. This cannot be rational.

      (iii) What property of choice is violated? Is there any utility function (strictly monotone in both goods) that can explain Dunno's choices?

      Solution: The weak axiom of revealed preference is violated. There can not be a utility function explaining all these choices.

[25%]

(b) Suppose that you learn that Dunno has the following binary relation as a preference over bundles:

$$(x_1, x_2)D(y_1, y_2) \Leftrightarrow (3x_1 + x_2 \geq 3y_1 + y_2 \text{ and } x_1x_2 \geq y_1y_2).$$

(i) Is Dunno's preference transitive?

Solution: Suppose that xDy and yDz. Then, we know that $3x_1 + x_2 \geq 3y_1 + y_2 \geq 3z_1 + z_2$ and $x_1x_2 \geq y_1y_2 \geq z_1z_2$ and it is evident that xDz. Transitivity is satisfied.

(ii) Is Dunno's preference reflexive, connected, complete?

Solution: It is apparent that D is reflexive. However, it is not connected because you can find different bundles that are not related, such as $(1, 1)$ and $(5, 0)$. Hence, it is not complete.

(iii) Is Dunno's preference antisymmetric?

Solution: It is not difficult to prove that the relation is not antisymmetric, because bundles $(1, 0)$ and $(0, 3)$ are indifferent. Sum is 3 and product is 0 in both cases.

[35%]

(c) Imagine that you discover that Dunno always goes to shop with some friend. At the shop, Dunno starts by eliminating all the bundles that are strictly worse than others that can be bought following the preference described in (b). If more than one bundle survives, Dunno freezes and has to ask one of his rational friends which of the remaining bundles to buy. They provide advise purely based on their own preferences.

(i) Explain why Dunno always eliminates the bundles that are not in the frontier of the budget constraint.

Solution: Interior bundles are dominated in both criteria, and thus worse than another alternative in the frontier. A formal argument may use (x_1, x_2) and some ϵ perturbation in any dimension.

(ii) Describe what bundles in the frontier could not be eliminated by Dunno in menus A_1 and A_2 described in (a) and conclude that Dunno froze in each of them.

Solution: Consider first menu A_1. Bundle $(10, 10)$ is better than any other

bundle in the second criterion. Only bundles with more of x_1 are better in the first criterion. It is clear that both criteria differ when moving over the frontier, so anything between $(10, 10)$ and $(20, 0)$ cannot be rejected.
Consider now menu A_2. The bundle that maximises the second criterion is $(7.5, 15)$ (equal expenditure). Anything between $(7.5, 15)$ and $(15, 0)$ could not be rejected.

(iii) Construct one menu A^*, defined by strictly positive income and prices, where you can prove that Dunno is able to select an alternative that is better than all the rest and not freeze.

Solution: The menu must be such that the point satisfying $p_1x_1 = p_2x_2$ in the frontier (and thus maximising the second criteria), is maximal for the first criterion. This can only be if prices are 3 and 1, so all points in the frontier are equally valuable for this criterion. Say, $3x_1 + x_2 \leq 30$. The most able students can prove that $(5, 15)$ is related to any other bundle, and the opposite cannot be true, though it is almost apparent with the reasoning.

(iv) Explain whether Dunno could have requested advise from the same rational friend in both days.

Solution: Because of rationality, we know that the same friend cannot give the advise. The most able students should prove something, because they would provide advise over very special subsets of the frontier (but the argument is the same).

[40%]

4. (a) In job 1, the wealth of an individual is $10 + \epsilon$, where $\epsilon \in [-10, 10]$ is a random shock with density function described by $f(\epsilon) = c(10 + \epsilon)^2$. In job 2, the wealth of an individual would be $10 + \mu$, where $\mu \in [-10, 10]$ is a uniform random shock (and thus, with density function described by $g(\mu) = \frac{1}{20}$).

(i) Show that $c = \frac{3}{8000}$. Sketch the two probability distribution functions (pdf) of wealth (or those of shocks, as you prefer).

Solution: It must be

$$\int_{-10}^{10} c(10 + \epsilon)^2 d\epsilon = \frac{c(10 + \epsilon)^3}{3}|_{-10}^{10} = \frac{8000c}{3} = 1$$

and the result follows. The pdf of the uniform is constant at $\frac{1}{20}$ from 0 to 20 (or from -10 to 10, depending on the choice). The pdf of job 1 is clearly convex, starting at 0 and ending at 1.

(ii) Consider any value $a \in [0,1]$. Show that, in job 1, the fraction of shocks that are worse than $20a^{\frac{1}{3}} - 10$ is equal to a.

Solution: We can write

$$\int_{-10}^{x} \frac{3}{8000}(10+\epsilon)^2 d\epsilon = \frac{\frac{3}{8000}(10+\epsilon)^3}{3}|_{-10}^{x} = \frac{(10+x)^3}{8000} = a$$

From here, the desired value of x must be $(8000a)^{\frac{1}{3}} - 10 = 20a^{\frac{1}{3}} - 10$.

(iii) In job 1, what is the fraction of wealth levels above 15?

Solution: The shock must be above 5. The corresponding value of a is given by $20a^{\frac{1}{3}} - 10 = 5$, i.e., $a = (\frac{3}{4})^3 = \frac{27}{64}$. Hence, $1 - \frac{27}{64} = \frac{37}{64}$ fraction of wealth levels are above 15.

[35%]

(b) Individual A is an expected utility maximiser, with Constant Relative Risk Aversion (CRRA) preferences. The coefficient of risk aversion is $r < 1$ and hence, we can simply write the utility function as w^{1-r}, where w is the final wealth of the individual.

(i) Is there any first-order stochastic domination between jobs 1 and 2? Denoting by CE_i the certainty equivalent of job i, explain how CE_1 and CE_2 are related, without computations.

Solution: It is evident from the analysis of the PDFs that the cumulative distribution function of job 1 is below that of job 2, i.e., job 1 first-order stochastically dominates job 2. Hence, every expected utility maximiser will prefer job 1 and the certainty equivalent of job 2 must be below that of job 1.

(ii) Denoting by EV_1 the expected value of wealth of job 1, show that $EV_1 = 15$. Is it always true that $CE_1 < 15$? Is it always true that $CE_1 < 10$?

Solution: For EV_1,

$$\int_{-10}^{10} \frac{3}{8000}(10+\epsilon)(10+\epsilon)^2 d\epsilon = \int_{-10}^{10} \frac{3}{8000}(10+\epsilon)^3 d\epsilon = \frac{3}{32000}(10+\epsilon)^4|_{-10}^{10} = \frac{240000}{32000} = 15$$

Since the individual is risk averse, it must be $CE_1 < EV_1 = 15$. However, the value can be above or below 10. Notice that the expected wealth is not 10 (it is just the middle point of the support) but 15, so an argument based upon risk aversion and the certainty equivalent below the expected value is wrong. It is then evident that if r is close to zero, the certainty equivalent would be close to 15 and hence above 10.

(iii) Compute the expected utility of job 1. Write down the equation that defines CE_1.

Solution: Simply

$$EU_2 = \int_{-10}^{10} \frac{3}{8000}(10+\epsilon)^{1-r}(10+\epsilon)^2 d\epsilon = \int_{-10}^{10} \frac{3}{8000}(10+\epsilon)^{3-r} d\epsilon$$
$$= \frac{1}{8000}\frac{1}{4-r}(10+\epsilon)^{4-r}|_{-10}^{10} = \frac{20^{4-r}}{8000(4-r)}$$

The certainty equivalent is the value CE_1 such that $(CE_1)^{1-r} = \frac{20^{4-r}}{8000(4-r)}$.

[35%]

(c) Consider all possible jobs that can be described as a wealth distribution in $[0, 20]$, like the two that you already analysed. Individual B has very strange preferences over these jobs. She only is interested in getting a wealth in between $(\underline{w}, \overline{w})$, where we know $\underline{w} < \overline{w}$. Hence, she prefers a job over another if and only if the fraction of wealth levels belonging to $(\underline{w}, \overline{w})$ is larger for the first job than for the second. Prove whether the following statements are true or false (if you cannot provide full proofs, intuitive arguments are also valuable).

(i) Individual B has complete and transitive preferences over jobs.

Solution: The first claim is true. Notice that we can write the utility function of a job as simply the fraction of wealth levels in the interval $(\underline{w}, \overline{w})$, and larger fractions correspond to better jobs. This represents the preferences.

(ii) Individual B prefers job 2 to job 1.

Solution: This is clearly not true. It is only the case for some intervals. For instance, if the interval $(\underline{w}, \overline{w})$ is contained in $(0, 10)$, it is evident that this individual prefers job 1 because more results can end up there.

[30%]

5. A risk-averse agent has a project that she must undertake. If the project is a success, then her final wealth will be 100; if it is a failure, then her final wealth will be 0. There are two possible effort levels that she can exert, low $e = 0$ or high $e = 1$: if $e = 0$, the probability of success is $\frac{2}{5}$, whereas if $e = 1$, the probability of success is $\frac{3}{5}$.

   The agent's utility from ending with wealth w having exerted effort e is $\sqrt{w} - e$, and she has no outside option.

   (a) Show that the agent will choose to exert high effort, and so her reservation utility is effectively 5.

   [20%]

   Solution: If $e = 1$, the expected utility is $\frac{3}{5}\sqrt{100} - 1 = 5$. If $e = 0$, the expected utility is $\frac{2}{5}\sqrt{100} = 4$, hence the agent chooses effort $e = 1$ and she can guarantee a utility of 5.

   Now assume that there is a single risk-neutral principal that can insure the agent, and that the success or failure of the project is public information.

   When effort is observable and contractible, the principal offers the agent a contract of the form: "If you exert effort e^*, then if the project succeeds you give me f, but if it fails I give you a c; if you exert any other effort level then we make no transfers."

   (b) What level of effort will the insurer choose? Which transfers will he specify? What are the expected profits of the insurer?

   [30%]

   Solution: The insurer can fully insure the agent, keeping her indifferent between accepting or no, so leaving a reservation utility of 5.
   For $e = 1$, the certainty equivalence of the agent is $\sqrt{CE_1} - 1 = 5$ so $CE_1 = 36$ which means that $f = 100 - 36 = 64$ and $c = 36$. The Principal gets an expected profit of $\frac{3}{5} * 64 - \frac{2}{5} * 36 = 24$.
   For $e = 0$, the certainty equivalence of the agent is $\sqrt{CE_0} = 5$ so $CE_0 = 25$ which means that $f = 100 - 25 = 75$ and $c = 25$. The Principal gets an expected profit of $\frac{2}{5} * 75 - \frac{3}{5} * 25 = 15$.
   Hence it is optimal to contract on $e = 1$.

   Some students might set up the whole problem: $\max_{\{f,c\}} \frac{3}{5}f - \frac{2}{5}c$ s.t. $\frac{3}{5}\sqrt{100 - f} + \frac{2}{5}\sqrt{c} - 1 \geq 5$. Solving this problem leads to the solution above but it is longer.

   When effort is non-observable, the principal offers the agent a contract of the form: "If the project succeeds you give me f, but if it fails I give you c."

   (c) What transfers would induce the agent to exert high effort? What contract will the insurer now offer to the agent?

   [40%]

Solution: The insurer needs to impose both IR and IC constraints. Denote by $v_H = \sqrt{100 - f}$ and $v_L = \sqrt{c}$, then:

$$IR : \frac{3}{5}v_H + \frac{2}{5}v_L - 1 \geq 5$$
$$IC : \frac{3}{5}v_H + \frac{2}{5}v_L - 1 \geq \frac{2}{5}v_H + \frac{3}{5}v_L$$

Some argument explaining or illustrating why the two constraints will bind is expected. These equation simplify to $3v_H + 2v_L = 30$ and $v_H - v_L = 5$ which leads to $v_H = 8$ and $v_L = 3$, hence $f = 36$ and $c = 9$.

The insurer payoff in this contract is $\frac{3}{5} * 36 - \frac{2}{5} * 9 = 18 > 15$, so it is still optimal to induce high effort.

(d) Comment briefly.
Solution: There is an agency cost of $24-18 = 6$ due to the effort being unobservable and the principal needing to provide incentives to the agent to exert effort and hence not fully insuring the agent. Any comment of the trade off between insurance and incentives will do. [10%]

6. A company owns a stockpile of recyclable paper that it processes and sells to paper mills. It operates over continuous time, $t \in [0, \infty)$. The size of the initial stockpile is $s(0) = B > 0$, which is falling at the rate of processing (cleaning, bundling), denoted by $x(t) \geq 0$ at time t. That is, $s'(t) = -x(t)$. The flow revenue is simply $x(t)$. The flow cost of production is $\frac{1}{2}x(t)^2$. Therefore the company's flow profit at time t is $x(t) - \frac{1}{2}x(t)^2$.

There are no other costs, for example, it is costless to store the remaining stockpile, and there are no fixed costs associated with processing the paper either. The company uses a discount rate $r > 0$ to discount flow profits.

(a) Denoting the remaining stock of recyclable paper by $s(t)$ at t, write down the company's problem of maximizing its discounted profits by processing and selling its stockpile over time as an optimal control problem in continuous time.
Solution: The problem is

$$\max \int_0^\infty e^{-rt} \left[x(t) - \tfrac{1}{2}x(t)^2\right] \, \mathrm{d}t$$
$$\text{s.t.} \quad s'(t) = -x(t)$$
$$s(t), x(t) \geq 0, \; s(0) = B.$$

(b) Using a Hamiltonian or otherwise characterize the optimal solution in terms of differential equations describing how x and s evolve over time. Sketch by hand or describe the corresponding phase diagram in the (s, x) plane, and explain.

Solution: The current value Hamiltonian is $H = x - \frac{1}{2}x^2 - \lambda x$ where λ is the co-state variable. The first-order conditions are

$$\begin{aligned} H'_x &= 0 \Leftrightarrow 1 - x - \lambda = 0, \\ -H'_s &= \dot{\lambda} - r\lambda \Leftrightarrow \dot{\lambda} - r\lambda = 0, \\ H'_\lambda &= \dot{s} \Leftrightarrow \dot{s} = -x. \end{aligned}$$

Differentiating the first equation in t we get $-\dot{x} - \dot{\lambda} = 0$. Substituting this as well as the original first equation into the second equation

$$\dot{x} = r(x-1).$$

This and $\dot{s} = -x$ constitute the ODE system.

Phase diagram in the (s, x) plane: $\dot{x} \geq 0$ iff $x \geq 1$, hence x is constant along the $x = 1$ horizontal line, increasing above it, decreasing below it. In contrast, s is decreasing everywhere, in proportion to x. A plot generated using random online software is below. Candidates are supposed to hand sketch this and explain, but can rely on software to get "inspiration".

(c) Verify whether the optimal processing schedule satisfies $x(t) = p + Ke^{rt}$ for some constant K.
Solution: Differentiating $x(t) = 1 + Ke^{rt}$ we get $x'(t) = rKe^{rt}$. Let's check the differential equation derived above,

$$x'(t) = rx(t) - r = r + rKe^{rt} - r = rKe^{rt},$$

yes, it indeed works. A good solution might add that K is determined by some boundary condition(s) – coming next.

(d) Assume that the company must get rid of its stockpile of recyclable paper within 1 unit of time, that is, by $t = 1$. (This is so because their site will be sold and they have to vacate it by then.) That is, $s(0) = B$ and $s(1) = 0$ by assumption. Find the optimal production schedule, $x(t)$, and interpret.
Solution: Integrate $s'(t) = -x(t) = -1 - Ke^{rt}$ to get

$$s(t) = -t - \tfrac{1}{r}Ke^{rt} + C,$$

where C is an integration constant.

The values of C and K are determined by the boundary conditions $s(0) = B$ and $s(1) = 0$:

$$s(0) = -\tfrac{1}{r}K + C = B, \ \ s(1) = -1 - \tfrac{1}{r}Ke^r + C = 0.$$

Express C from the latter and sub into the former:

$$B = 1 + \tfrac{1}{r}K(e^r - 1),$$

hence

$$K = \frac{r(B-1)}{e^r - 1}.$$

The optimal schedule is

$$x(t) = 1 + \frac{re^{rt}}{e^r - 1}(B-1).$$

In the phase diagram: depending on the sign of $B-1$ we might hop on a path above or below the $x = 1$ line in order to hit the vertical axis by $t = 1$. If $B = 1$ then $x(t) = 1 = B$, that is, the company processes the pile linearly, $s(t) = (1-t)B$ for $t \in [0, 1]$.